INDUSTRY STARTER PACKS FOR ENTERPRISE ARCHITECTS

Pharmaceuticals

The complete set — what is in it, what each piece is for, and the order to pick them up

Two volumes, six spreadsheets, three decks and the worked example they were all built against. Read this first: it says which file answers which question, and which three you need in your first week.

South African lens. Sourced throughout. October 2026.

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Thotnr Consulting

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GUIDE 1

What is in this set

Thirteen files. Three of them matter in your first week.

This set exists so that an architect who has never worked in pharmaceuticals can walk into one and be useful. Not eventually — within a quarter.

It comes in four parts and a worked example. The two books teach and instruct, the spreadsheets are what you actually fill in, the decks are what you put on a screen in front of other people, and the Calderon files are a complete, populated example of what the finished thing looks like — defects and all.

Read these three first

> **IF YOU READ NOTHING ELSE**
>
> **`01 Learner guide`** — read it once, front to back, before you start. It is the industry.
>
> **`02 Workbook`** — never read front to back. Open it at the phase you are in and work the numbered steps.
>
> **`03 Excel templates / Pharmaceuticals_EA_Metamodel_TEMPLATE.xlsx`** — the empty repository. Everything the workbook tells you to do lands in it.

The whole set

Folder	File	What it is	When you use it
01	Pharmaceuticals — a starter pack for enterprise architects	The learner guide. 246 pages, 23 items, 118 figures.	Read once, keep beside you
02	Pharmaceuticals — the architecture workbook	The ADM workbook. 138 numbered steps, 84 printable forms, 19 question sets.	Every working day
03	Pharmaceuticals_EA_Metamodel_TEMPLATE.xlsx	The empty repository. 115 sheets, with change control, a computed index and the four sheets the example lacks.	From day one
03	Pharmaceuticals_Architecture_Comparison_Tool_TEMPLATE.xlsx	Baseline against target, with a validation-impact column on every difference.	Phase E, then quarterly
03	Pharmaceuticals_EA_Workshop_Capture.xlsx	Live capture for the sessions that produce rows.	In the room
03	Pharmaceuticals_EA_Transcription_Forms.xlsx	47 forms matching the printed ones.	Typing up paper
03	Pharmaceuticals_EA_Principles_and_Standards.xlsx	33 live principles and 25 standards, ready to adopt or argue with.	Preliminary phase
03	Pharmaceuticals_EA_Maturity_Assessment.xlsx	35 scored questions across 11 domains, plus a register of 7 absences.	First month, then annually
04	Pharmaceuticals_EA_Executive_Briefing.pptx	31 slides for the executive committee.	End of the first quarter
04	Pharmaceuticals_EA_View_Templates.pptx	49 slides: 36 ArchiMate 4 views, drawn and guided.	Whenever you draw
04	Pharmaceuticals_EA_Workshop_Pack.pptx	55 slides facilitating eight workshops.	In the room
05	Pharmaceuticals_TOGAF_Metamodel.xlsx	The worked example — a populated repository for an invented company.	When you want to see what finished looks like
05	Pharmaceuticals_ArchiMate4_Views.pptx	Worked ArchiMate 4 views of that repository.	Alongside the view templates

The two books, and why they are separate

	Volume 1 — the learner guide	Volume 2 — the workbook
Read	once, front to back	never front to back
Answers	what this industry is and how it runs	what do I do on Monday
Shelf life	keep it as reference	consumed — you write in it

Volume 1 covers the eight archetypes, the vocabulary, the business model, who does what, how pharmaceutical companies make money, the operating model and the portfolio cadence, the process model in BPMN, the conceptual data and information models, the application landscape, GxP and validation, the reference architectures such as they are, the providers, the regulator and the inspector, and how to model all of it in ArchiMate 4. Volume 2 takes the ADM phase by phase and says which sheet, which columns, who signs it, how you know it is done, and how it usually goes wrong.

They interlock. Every step in volume 2 with teaching behind it carries a **Look it up** line naming the volume 1 chapter and section, and volume 2's appendix E indexes the whole workbook back to volume 1.

The chapter no other volume in this series has

Volume 1 chapter 13 is **GxP, validation and data integrity**. Banking has no equivalent. Nor does insurance, retail or manufacturing. In every other industry the architecture governs the systems; here the systems are themselves regulated. A change to a computerised system that holds a GxP record is a validation event before it is a backlog item, validation is a state you maintain rather than a thing you did, and "just patch it on Tuesday" does not exist.

That is the chapter to read twice, and volume 2's Phase D, Phase G and Phase H chapters are its working halves.

Print volume 2

Volume 2 is built to be printed and written in: 84 printable forms whose columns are the real columns of the real sheets, 19 question sets with ruled answer space, and working tables inside the steps.

> **THE TRANSCRIPTION LOOP**
>
> Paper in the room, spreadsheet the same afternoon, repository record with a change reference. Paper that is never typed up is a wasted workshop; a record typed up without a change reference is a repository that will rot inside a year.

The two invented companies, kept apart

Veldtkruin Group is volume 1's worked example: Veldtkruin Pharmaceuticals, a generics manufacturer with an oral solid dose site at Isando and a sterile fill-and-finish site at Brackenfell; and Khanyisa Bio-Development, a contract developer and manufacturer at Stellenbosch. It exists so that the teaching can be concrete.

Calderon Therapeutics is the organisation in the populated repository in folder 05. It exists so that you can see a finished repository and practise on it.

Both are inventions. Neither describes any real organisation, and their data is never mixed. Every figure in either is illustrative and internally consistent rather than sourced.

The worked example is deliberately imperfect

The repository in folder 05 is populated, complete-looking and wrong in about thirty places. That is the point. It declares six industry object types — compound, study, batch, regulatory submission, GxP computerised system, adverse event — and holds no instances of any of them. Eight of its fourteen data entities name the wrong master system, and they are wrong by rotation rather than by guesswork. Its subject data is classified `Internal`. Nobody in it does a pharmaceutical job. Ten of its twenty-five relationship sheets are index-order joins rather than statements about the world. Its dashboard claims 850 records over 814.

Volume 2 works through every one of those, because a repository that looks finished and is not true is the condition you will actually inherit.

The notation

ArchiMate 4 throughout for architecture, BPMN 2.0 only for level-2 process models where the hand-off is the point. Three rules the volumes keep: the element is `Role`, with no layer-prefixed variant; there is no Constraint element, so a constraint is expressed as a `Requirement`; and there is no metric element and `Value` is not a number, so measures live in a measure definition register and never inside the model.

On sourcing

Everything factual in both volumes carries a citation with a retrieval date, and where a figure could not be sourced the text says so rather than estimating. Standards versions move, so every version in these files is stated with the date it was read and an instruction to re-check before it reaches a client deliverable. Treat the dates as the shelf life of the claim.